

Table 1: Selected Hyperparameters for Lapse Prediction Models

Model	Selected hyperparameters
Logistic Regression	$C = 9.2850$ penalty = L1 solver = saga max_iter = 1000 tol = 10^{-4}
Decision Tree	max_depth = 10 min_samples_split = 167 min_samples_leaf = 26 criterion = gini
Random Forest	n_estimators = 285 max_depth = 8 min_samples_split = 115 min_samples_leaf = 48 max_features = sqrt subsample = 0.7047 criterion = entropy
XGBoost	n_estimators = 598 learning_rate = 0.1388 max_depth = 9 min_child_weight = 20 subsample = 0.6795 colsample_bytree = 0.8731 reg_alpha (L1) = 0.5394 reg_lambda (L2) = 0.1247
LightGBM (Full)	n_estimators = 471 learning_rate = 0.1401 num_leaves = 135 max_depth = 9 min_child_samples = 69 subsample = 0.9977 colsample_bytree = 0.5589 reg_alpha (L1) = 2.69×10^{-6} reg_lambda (L2) = 3.24×10^{-5}
LightGBM (98%)	n_estimators = 499 learning_rate = 0.0669 num_leaves = 32 max_depth = 7 min_child_samples = 98 subsample = 0.8108 colsample_bytree = 0.5044 reg_alpha (L1) = 0.3526 reg_lambda (L2) = 0.3953
LightGBM (90%)	n_estimators = 249 learning_rate = 0.1080 num_leaves = 27 max_depth = 6 min_child_samples = 79 subsample = 0.7750 colsample_bytree = 0.6419 reg_alpha (L1) = 0.1677 reg_lambda (L2) = 1.0729
LightGBM (80%)	n_estimators = 300 learning_rate = 0.0669 num_leaves = 32 max_depth = 7 min_child_samples = 98 subsample = 0.8108 colsample_bytree = 0.5044 reg_alpha (L1) = 0.3526 reg_lambda (L2) = 0.3953
Multi-Layer Perceptron (MLP)	hidden_layer_sizes = 224 activation = tanh alpha (L2) = 7.18×10^{-5} learning_rate_init = 0.000738 batch_size = 64 max_iter = 500